

LegalEase: AI-Powered Legal Document Generator

Final Project Report

Project Phase	7. Project Documentation
Team	Team 3
Date	02 October 2026
Prepared From	Supplied LegalEase repository + supplied project templates

1. Project description

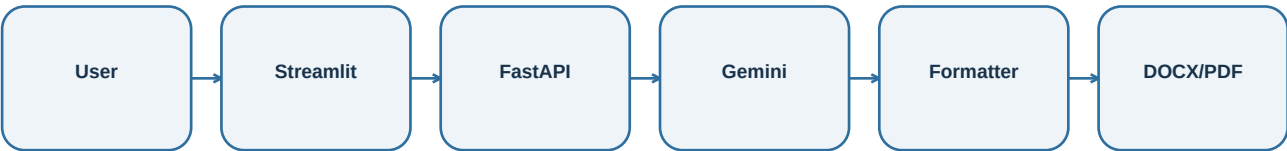
LegalEase is an AI-powered legal document drafting application combining Streamlit, FastAPI and Google Gemini. Users select a document type, provide parties, terms and dates, review the generated draft and download DOCX/PDF.

2. Use cases

Common-document first drafts including NDAs, employment contracts, lease agreements, service agreements, partnership agreements and custom document types.

3. Technical architecture

High-level flow



4. Prerequisites

Python 3.10+, requirements.txt dependencies, a Gemini API key and local ports for FastAPI and Streamlit.

5. Prior knowledge

Python, REST APIs, FastAPI, Streamlit, environment variables, prompt engineering and document generation.

6. Objectives

Objective	Status
Guided legal document inputs	Implemented
AI-generated structured draft	Implemented
Preview	Implemented
DOCX/PDF export	Implemented

Objective	Status
Validation/error handling	Implemented
Persistent history / production deployment	Future / not in supplied repository

7. Workflow

User → Streamlit form → POST /generate → FastAPI validation → Gemini prompt → generated text → preview → DOCX/PDF → download.

8. Milestones

Requirements → setup → core AI/API → frontend/export → testing → documentation/demo.

9. Results

Automated tests cover validation, mocked successful generation, Gemini failure handling, sanitization, DOCX/PDF formatting and the root endpoint.

10. Advantages

Simple architecture, lightweight UI, configurable Gemini model, document-type guidance, immediate preview and two export formats.

11. Limitations

AI output can be inaccurate; no persistent history/database; no production deployment/HA; no semantic legal verification; performance benchmarking is not established.

12. Future enhancements

Jurisdiction-aware clauses, citations, human/legal review, document history, authentication, persistent storage, audit logs, structured validation, deployment automation and observability.

13. Conclusion

LegalEase demonstrates an end-to-end AI drafting workflow. The supplied repository establishes the generation, API, UI, formatting and testing layers; remaining work is production hardening, validation, deployment and governance.

Source basis: Implemented claims are based on the supplied LegalEase repository. Items without repository evidence are explicitly marked pending, future, or not implemented.